PharmAchieve

PARKINSON’S DISEASE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care: Clinical Care	Providing Care: Distribution	Knowledge & Expertise
Communication & Collaboration	Leadership & Stewardship	Professionalism

LEGEND

3-MT	3-Methoxytyramine
3-O-MD	3-O-Methyldopa
BBB	Blood Brain Barrier
BID	Bis In Die (Twice Daily)
BP	Blood Pressure
BPM	Beats Per Minute
BPH	Benign Prostatic Hyperplasia
CBC	Complete Blood Count
CI	Contraindication
COMT	Catechol-O-Methyl Transferase
CR	Controlled Release
CYP	Cytochrome P450
DDCi	DOPA Decarboxylase Inhibitor
ESR	Erythrocyte Sedimentation Rate
HR	Heart Rate
HTN	Hypertension
IR	Immediate Release
L-dopa	Levodopa
MAO-B	Monoamine Oxidase Type B
MOA	Mechanism of Action
NMDA	N-Methyl-D-Aspartate
N/V/D	Nausea and/or Vomiting and/or Diarrhea
OT	Occupational Therapy
PD	Parkinson’s Disease
PO	Per Os (Oral)
PRN	Pro Re Nata (As Needed)
PT	Physiotherapy
QAM	Quaque Ante Meridiem (Every Morning)
QHS	Quaque Hora Somni (Every Night At Bedtime)
QID	Quarter In Die (Four Times Daily)
S/E	Side Effects
SLP	Speech-Language Pathologist
SNRI	Serotonin-Norepinephrine Reuptake Inhibitor
SSRI	Selective Serotonin Reuptake Inhibitor
T2DM	Type 2 Diabetes Mellitus
TID	Ter In Die (Three Times Daily)
TCA	Tricyclic Antidepressant

WHAT IS PARKINSON’S DISEASE?

Parkinson’s Disease

A chronic neurodegenerative and heterogenous disorder with clinical presentations that vary from patient to patient

Cause

- Partly caused by the progressive loss of dopaminergic neurons in the substantia nigra
  - Loss of these neurons causes a deficit of dopamine which leads to tremor, rigidity, akinesia, and bradykinesia
- Degradation of non-dopaminergic neurons may also be a cause of many symptoms of Parkinson’s Disease, but further investigation is needed to establish significance

PATHOPHYSIOLOGY

FYI

- Onset of symptoms does not occur until significant loss of dopaminergic production in the substantia nigra
- Affects the extrapyramidal system which controls voluntary movement

Loss of dopaminergic neurons may be caused by:

- Oxidative stress/damage
- Environmental toxins
- Genetic predisposition
- Accelerated aging

RISK FACTORS

Age 65+
Advanced age is most important risk factor

Males affected 2:1

Genetics

Metabolic Syndrome

T2DM

Traumatic Brain Injury

History of melanoma or prostate cancer

Pesticide exposure

Air pollution

High consumption of dairy products

Low vitamin D

Use of well water

CLINICAL PRESENTATION AND COMPLICATIONS

T — TREMOR

- Most common presenting symptom
- E.g. resting, postural, pill-rolling

R — RIGIDITY

- Clinical hallmark
- Increased resistance to passive motion
- 2 types: cogwheel & lead-pipe

A — AKINESIA/ BRADYKINESIA

- Most disabling motor feature
- Akinesia: absence or lack of movement
- Bradykinesia: slowness and difficulty in maintaining movement

P — POSTURAL INSTABILITY

- Least specific, most disabling
- Increases risk of falls
- "Phantom pillow syndrome"
- Not responsive to drug treatment

CLINICAL PRESENTATION AND COMPLICATIONS

<table>
<tr><td rowspan="2">Motor Symptoms</td><td colspan="3">Non-motor Symptoms</td></tr>
<tr><td>Emotional</td><td>Cognitive</td><td>Autonomic</td></tr>
<tr><td>Decreased manual dexterity
Slurred speech (dysarthria)
Difficulty swallowing (dysphagia)
Shuffling gait (festination)
Reduced facial expression (hypomimia)
Weak voice amplification (hypophonia)
“Freezing” at initiation of movement
Cramped handwriting (micrographia)</td><td>Apathy
Depression
Anxiety</td><td>Impaired executive function
Speech and language
Memory
Comprehension
Hallucinations
Delusions</td><td>Urinary and fecal incontinence
Constipation
Sleep disturbances
Orthostatic BP changes
Sexual dysfunction
Drooling (sialorrhea)
Sweating (diaphoresis)</td></tr>
</table>

DIAGNOSIS

Suspect PD in anyone presenting with tremor, gait disorder, balance problems, stiffness (rigidity), slowness (bradykinesia)*

↓

Rule out secondary causes (e.g. stroke, trauma, infections, drug causes, dementia)

↓

Consider a dopamine replacement therapy trial in patients with possible PD, coupled with regular long-term follow ups

↓

Regular follow-ups to review the diagnosis

Diagnosis is based predominantly on clinical features. No lab tests are available to diagnose PD

PD can be diagnosed using the Movement Disorder Society (MDS) clinical diagnostic criteria

* As per MDS, PD is defined as bradykinesia plus rest tremor or rigidity

DRUG-INDUCED PARKINSONISM

Parkinson's disease	Drug-induced parkinsonism
Asymmetrical (one side)	Bilateral (both sides), symmetric
Does NOT start or stop when causative agent is started or discontinued	Symptoms start when the causative agent is started, and stop when the agent is discontinued
Responds well to levodopa	Does not respond to levodopa

Must be ruled out in suspected Parkinson's Disease

Cause

- Antipsychotics → act as dopamine antagonists
- Antiemetics such as metoclopramide, prochlorperazine block dopamine centrally and peripherally
- Valproic acid, lithium → side effect of tremor may be mistaken as PD
- TCAs and SSRIs have case reports of abnormal movement disorders (mechanism unclear)

Treatment

If possible: discontinue causative drug, reduce dose, or switch therapies. If drug can't be altered, consider adding anticholinergics or amantadine. Re-assess every few months.

GOALS OF THERAPY

- ✓ Improve symptoms, PD has no cure
- ✓ Decrease disability with daily living
- ✓ Manage symptoms with medications (no cure)
- ✓ Prevent and treat complications associated with medications
- ✓ Prevent and treat complications related to PD
- ✓ Identify possible drug-induced complications
- ✓ Improve quality of life

NON-PHARMACOLOGICAL MEASURES

Inadequate to properly manage PD

- Pharmacologic therapy is required

Early Stages of Disease

- Patient and caregiver education should be provided
- Encourage activity and exercise as tolerated
  - Aerobic, strength, balance, and gait exercises are all good options
    - Tai chi, yoga, dancing
- Utilize allied health professionals depending on symptoms
  - E.g. SLPs, OT, PT, dieticians

Surgery

- Not curative
- Functioning will increase to be as best "on" times, does not return to baseline prior to PD
- No postural instability improvement
- Contraindication: poor cognition

PHARMACOLOGICAL MEASURES

Initiation of Therapy

- Generally, treatment is initiated once the disease begins to interfere with any of the following:
  - Activities of daily living
  - Employment
  - Quality of life
- Patients do not have to begin treatment right at the start of symptoms if they are not bothered by them

Before Starting

- Discussion should be had with patient regarding their individual case
  - Clinical circumstances, age, symptoms, comorbidities, and risks from polypharmacy
  - Lifestyle, preferences, needs, and goals
  - Potential benefits and harms of the drug classes used in treatment

PHARMACOLOGICAL MEASURES

OVERVIEW

MAO-B Inhibitors

- Selegiline
- Rasagiline
- Safinamide

Anticholinergics

- Benztropine
- Ethopropazine
- Trihexyphenidyl

NMDA Antagonist

- Amantadine

COMT Inhibitor

- Entacapone

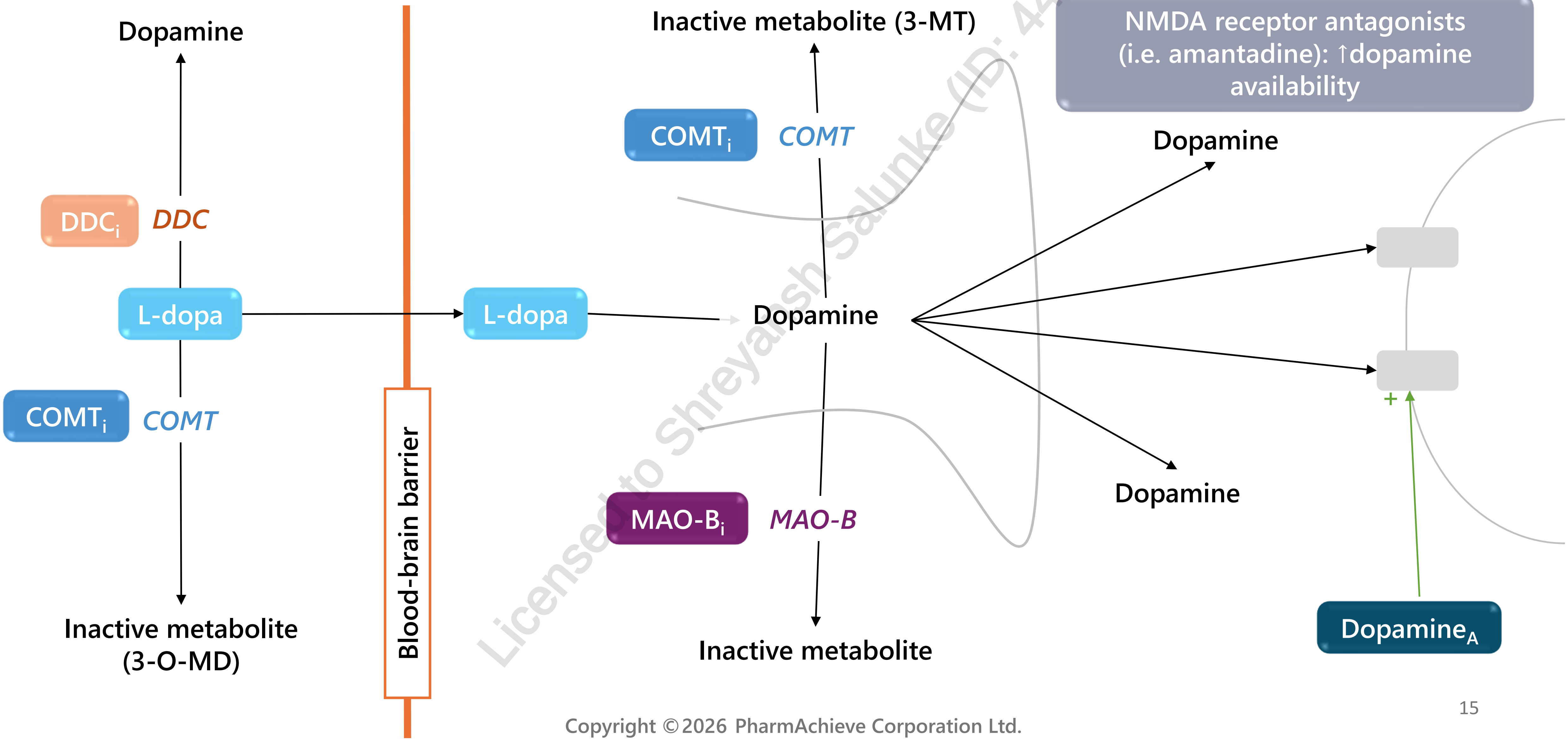
Dopamine Agonists

- Bromocriptine
- Ropinirole
- Pramipexole
- Rotigotine

Dopamine Precursor + DOPA Decarboxylase Inhibitor Combinations

- Levodopa + Carbidopa
- Levodopa + Benserazide

DOPAMINE METABOLISM



PHARMACOLOGICAL MEASURES

OVERVIEW

Motor Symptoms in Early Disease (choose one)

- Dopamine precursor (L-Dopa) + DOPA decarboxylase inhibitor
  - Most effective for motor symptoms
- MAO-B inhibitors
  - Used as monotherapy for symptomatic treatment in early PD
- Non-ergot dopamine agonists

Motor Symptoms in Later Disease (combination therapy)

- Combination therapy is necessary
- COMT inhibitors may be added in advanced disease
- Amantadine can be used in later stages

MAO-B INHIBITORS - SELEGILINE, RASAGILINE, SAFINAMIDE

- MOA: Selective MAO-B inhibitors inhibiting CNS dopamine degradation = ↑ dopamine levels
- Bind irreversibly to MAO-B and may be used as monotherapy for early PD or as adjunctive therapy
- Safinamide: newest MAO-B inhibitor, not used as monotherapy, usually used as adjunct with levodopa

Rasagiline
• More potent; can be used initially to improve motor symptoms and, in advanced disease, for "wearing-off" symptoms
• Longer duration of action: once daily dosing
• Selective MAO-B inhibitor: not associated with interactions involving tyramine (however large amounts of tyramine should be avoided)
• CYP1A2 substrate: dose adjust with CYP1A2 inhibitors (e.g. ciprofloxacin)
• Coadministration with high-fat meals can decrease bioavailability

Selegiline
• Shorter duration of action: BID dosing
• MAO-B selectivity is lost at high concentrations
• Has an amphetamine metabolite: avoid sympathomimetics (e.g. pseudoephedrine) and avoid taking at night
• Selegiline-specific side effects: sympathomimetic effects (due to interactions involving tyramine): HTN, tachycardia, jitteriness

Precautions:

- Serotonin syndrome possible when taken with other serotonergic drugs
- Can interact with tyramine-containing foods If given at doses above the maximum daily dose → hypertensive crisis
- Adverse effects: nausea, headaches, orthostatic hypotension, insomnia, serotonin syndrome

ANTICHOLINERGICS

- MOA: Counteract increased cholinergic activity (↓ dopamine) which contributes to tremor
- NOT first line → limited efficacy & many side effects
  - Trihexyphenidyl (taken TID), benztropine (taken QHS), ethopropazine (taken TID)

Pearls	• May be considered in young patients with early PD and prominent tremor • Effective against tremor but no effects on bradykinesia

Precautions	• Not recommended in elderly patients (can cause confusion and memory difficulties) • Should not be used in patients with dyskinesias and/or motor fluctuations • C/I: glaucoma, BPH, patients with dementia (due to cognitive impairment) • Adverse effects: confusion, memory impairment, hallucinations, dry mouth, blurred vision, urinary retention, somnolence, constipation

NMDA ANTAGONIST- AMANTADINE

- MOA: Poorly understood, some postulates:
  - Antiviral with anticholinergic & anti-glutamate properties = blocks acetylcholine receptor
  - May release dopamine from presynaptic terminals or block its reuptake
- Effects are mild-moderate and temporary

Benefits	• Useful in later stages to reduce L-dopa induced dyskinesia (choreic movement) • Easy to use and well-tolerated • Fixed dosing of 100 mg BID or TID • Can interfere with sleep, avoid administration later in the day
Precautions	• Insufficient evidence for use in early PD • Adjust dose for renal dysfunction • Rebound symptoms with discontinuation so taper slowly • Caution: patients with cognitive deficits, can increase confusion • Adverse effects: hallucinations, confusion, nightmares, insomnia, leg edema, nausea, dry mouth, blurred vision, livedo reticularis (diffuse mottling skin)

DOPAMINE AGONISTS

- MOA: Activate dopamine receptors by mimicking dopamine

Benefits	• Used monotherapy as symptomatic treatment for people with early PD and/or as adjunct with levodopa in advanced motor complications • Second most potent class of medication after levodopa for control of motor symptoms of PD • Prolongs the effective treatment period in patients with deteriorating response ("off" time reduced) • Used in patients who cannot tolerate high-dose L-dopa (allows for reduction of L-dopa dose) • Compared to levodopa – less effective in controlling motor symptoms but less likely to cause fluctuations in early disease

Ergot	Non-Ergot
• Bromocriptine • NOT first line agent, avoid	• Pramipexole, Ropinirole, Rotigotine • Do not carry risk of ergots and are safer • Similar efficacy profiles of all 3
• Prior to Bromocriptine initiation, obtain baseline erythrocyte sedimentation rate (ESR), renal function, cardiac echocardiogram and chest x-ray and annually due to risk of pleuropulmonary and cardiac valve fibrosis	• Rotigotine: transdermal patch = reduce PO meds + improve adherence + stable plasma concentrations to avoid pulsatile stimulation of dopamine receptors. Change patch every 24 hours but should not place in the same area of the skin for 14 days

DOPAMINE AGONISTS

Higher incidence of side effects compared to levodopa

Adverse effects

- Somnolence, excessive daytime sleepiness, sleep attacks
- Nausea/vomiting
- Orthostatic hypotension
- Nightmares
- Hallucinations
- Confusion
- Dizziness
- Dyskinesia
- Edema
- Compulsive behaviours
  - Gambling, shopping, hypersexuality
- Pulmonary fibrosis (bromocriptine)

Additional Considerations

- Titrate dose slowly over 4-6 weeks to decrease side effects of nausea, light-headedness, sleepiness and hallucinations
- Caution/Avoid in patients > 70 years

COMT INHIBITORS (ENTACAPONE)

- MOA: Prevent the breakdown of L-dopa in the periphery and enhance central L-dopa bioavailability → increase and prolong levodopa effects

- Has no effect if not used in conjunction with L-dopa
  - Given with each dose of L-DOPA to prolong half-life (by 1-2.5 hours) and prevent 'wearing off'
  - Available in combination with levodopa/carbidopa (Stalevo®)

- Decrease Levodopa dose (by 10 – 30%) when initiating COMT inhibitors to reduce risk of dopaminergic side effects (e.g dyskinesia, psychosis, diarrhea, and abdominal pain)

- May cause brownish-orange urinary discoloration

LEVODOPA

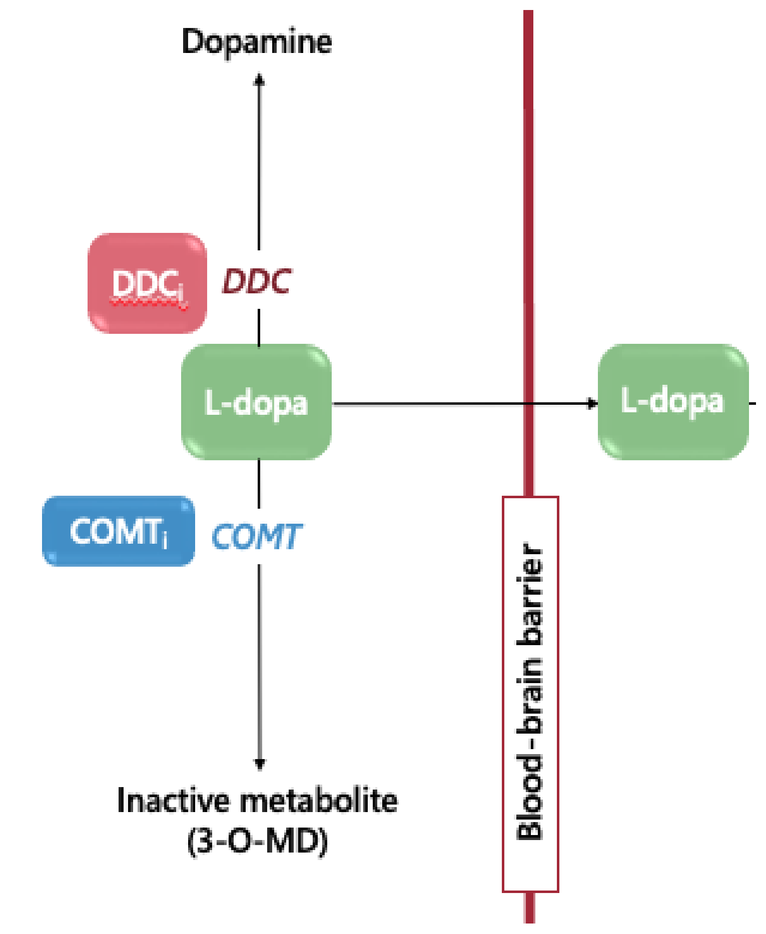
Gold standard of Parkinson's Treatment

- Eventually, all patients will need to use Levodopa

MOA

- L-dopa is a precursor to dopamine
  - Used since peripheral dopamine cannot cross BBB
- Given with a DDCi (carbidopa or benserazide) to inhibit peripheral transformation to dopamine, allowing more L-dopa to enter the brain
- The DDCi allows lower doses of L-dopa to be used, which decreases dose-dependent side effects such as nausea, and vomiting

Examples	• Levodopa/carbidopa IR, CR version also available • Levodopa/benserazide (Prolopa®) • Levodopa/carbidopa intestinal gel (Duodopa®)



LEVODOPA

Benefits	• Most effective at treating motor symptoms • Can be used for symptom management in early disease

Precautions	• Associated with earlier onset of
	  • Peak-dose effect dyskinesias
	  • “Wearing-off” motor fluctuations
	• Delayed- or no- “on” response may occur with CR formulations
	• Dietary protein and iron interfere with levodopa absorption (↓ bioavailability)

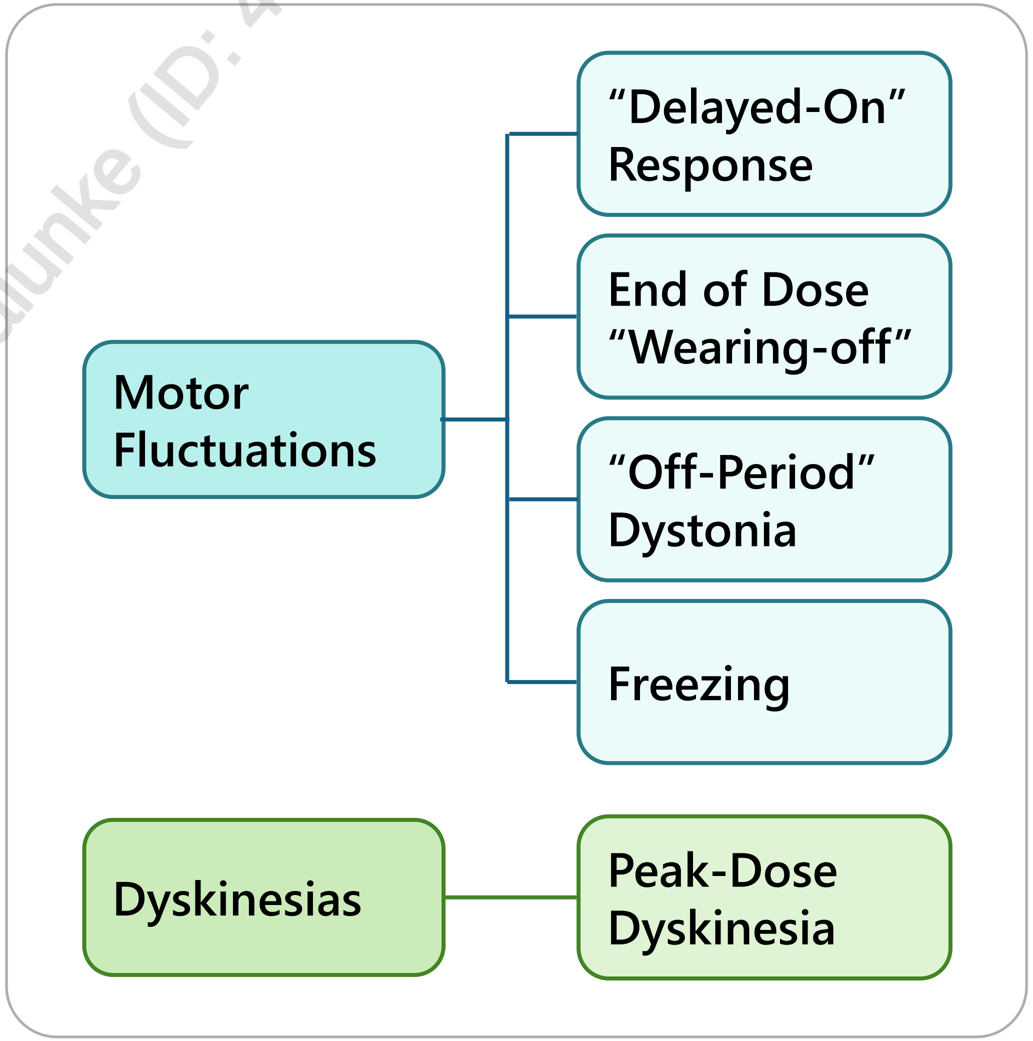
Adverse Effects	• N/V/D
	• Orthostatic hypotension
	• Cardiac arrhythmias
	• Restlessness
	• Dyskinesia
	• Complications related to percutaneous intrajejunal tube placement
	  • Obstructions
	  • Infection

LEVODOPA COMPLICATIONS

OVERVIEW

- In the early stages of PD, patients enjoy a long-lasting response following a single dose of levodopa. With disease progression and longer-term treatment, L-dopa complications can arise
  - These complications develop in up to 50% of patients after 5 years

- Pathophysiology of motor complications is unclear; leading theory suggests they are related to pulsatile stimulation of dopamine receptors resulting from intermittent dosing and levodopa's short half-life



MANAGING LEVODOPA COMPLICATIONS - “DELAYED-ON” RESPONSE

Delayed gastric emptying or decreased absorption in the duodenum results in slower onset

Chew or crush tablets, as appropriate, and drink a full glass of water

Use regular tablet formulation on empty stomach to facilitate gastric emptying

Avoid taking with protein and iron (if concerns with delayed onset or variable effect, may advise to take 30-60 minutes before protein-rich meals)

MANAGING LEVODOPA COMPLICATIONS - END OF DOSE "WEARING-OFF"

Occurs due to advancing disease and results in return of tremor and motor rigidity

- Duration of action of levodopa shortens due to dopamine neuronal storage capacity and short half-life

Increase levodopa dose and/or frequency

Add dopamine agonist

Add COMT inhibitor

Add MAO-B inhibitor

Bedtime CR formulation administration

Bedtime dose of a dopamine agonist

CR regimen may improve wearing-off and nighttime akinesia

Intrajejunal levodopa/carbidopa

Bedtime dosing may help to reduce nocturnal episodes and improve functioning upon awakening

MANAGING LEVODOPA COMPLICATIONS - “OFF-PERIOD” DYSTONIA

“Off-period” dystonia is defined as sustained muscle contractions, usually occurring in the early morning, because of waning drug levels

- Bedtime CR formulation dose
- Consider COMT inhibitor*
- Consider MAO-B inhibitor*

* in advanced PD with motor fluctuation

MANAGING LEVODOPA COMPLICATIONS - FREEZING

Freezing is generally the sudden inhibition of movement, but it may affect other body parts, even speech

- There are no pharmacologic treatments or medication adjustments to help with freezing

Physiotherapy

4 “S” strategy

- Stop
- Sigh
- Shift
- Step

Other non-pharmacologic measures

MANAGING DISEASE COMPLICATIONS - PEAK-DOSE DYSKINESIAS

Involuntary choreiform (dance-like) movements secondary to dopamine receptor over-stimulation, which occurs during the peak effect of the dose (~1 – 2 hours after administration)

- Decrease dose by a small amount and give more frequently
- Decrease dose, and add or increase dopamine agonist
- If on MAO-B, taper off and stop medication
- If COMT inhibitor recently added, decrease levodopa dose by 10-30%
- Add amantadine

LEVODOPA – TIPS

IR formulations have a faster onset, therefore, are more useful for "delayed-on" in the morning

CR has erratic absorption and is less absorbed than IR despite having similar side effects

Take with food early in therapy to help with nausea

Separate from protein-rich foods/meals due to decreased bioavailability

Takes weeks to reach full efficacy at target dose

Keep levodopa dose as low as possible to prevent motor complications

MANAGING DISEASE COMPLICATIONS

	Nausea/ Vomiting	• OK to use domperidone (peripheral dopamine antagonist) • Avoid metoclopramide because it crosses BBB
	Constipation	• Increase fiber & fluid intake, laxatives • Increase physical activity • Reduce or stop anticholinergic drugs • Add domperidone
	Depression	• Avoid SSRI, SNRI, and bupropion if on MAO-B inhibitor due to increased risk of serotonin syndrome

MANAGING DISEASE COMPLICATIONS - DRUG-INDUCED PSYCHOSIS

- Monitor electrolytes, look for signs of infection (if acute presentation)
- Simplify regimen by discontinuing drugs in the following order:
  - Stop: anticholinergics, TCAs, antihistamines, anxiolytics, sedatives
  - Taper & stop: amantadine
  - Taper & stop or reduce: Dopamine agonist
  - Taper & stop or reduce: MAO-B inhibitors, COMT inhibitors
  - Reduce: L-dopa
- Add atypical antipsychotic (quetiapine, clozapine preferred but requires monitoring) if disruptive hallucinations and psychosis persists
  - Avoid all other antipsychotics
  - Olanzapine, risperidone, and aripiprazole can worsen parkinsonism

MONITORING PARAMETERS

Parameter	Degree of Change	Timeframe
Dyskinesias, freezing, “wearing-off” symptoms	Minimize	Ongoing
Non-motor symptoms	Improve or prevent decrease	
Falls, orthostatic hypotension	Prevent	
Balance	Improve	
Morning stiffness	Improve and prevent	
Side effects from medications	Prevent or minimize	

THERAPEUTIC TIPS

Patient & family education about disease progression is key

Treatment should be based on whether patient is bothered by symptoms

Educate patients about when symptoms/side effects appear in relation to dose (dyskinesia, “wearing-off” effect)

Ask patient when and how medications are taken as this will impact efficacy of the drug → important for L-dopa since food (especially protein) decreases bioavailability

Complications of L-dopa include motor fluctuations, dyskinesia, predictable “wearing-off” effect, on/off state

Eventually all patients will be on L-dopa (most effective but has most dopaminergic side effects)

Avoid abrupt cessation of antiparkinsonian medications (risk of parkinsonism-hyperpyrexia syndrome [similar to neuroleptic malignant syndrome] or acute akinesia)

TAKEAWAY

- PD is partly caused by loss of dopaminergic neurons → rule out drug-induced causes
- Levodopa is used as initial therapy for patients regardless of age
  - Take separately from protein-rich meals as this can affect bioavailability
- Treatment:

Early PD	Later stages of PD
MAO-B inhibitor, or non-ergot dopamine agonist, or L-Dopa + DOPA decarboxylase inhibitor → depending on patient-specific factors and goals	Combination therapy – may add amantadine, or entacapone (must already be on L-Dopa + DOPA decarboxylase inhibitor)

- End-of-dose/wearing-off: may consider adding COMT inhibitor (entacapone) as it can prolong levodopa's half-life
  - Decrease levodopa's dose to avoid peak-dose dyskinesias
- Domperidone for N/V, avoid metoclopramide due to its central activity
- Approach drug-related psychosis by sequentially discontinuing certain drugs
- COMT inhibitors can discolour urine
- Dopamine agonists can cause impulse disorders – need to educate patient and caregiver on risks

CASE 1

DH is a 72-year-old male who presents to your pharmacy concerned about the tremors he has been experiencing recently. DH presents to the family medicine clinic with his wife. He complains of progressively worsening left arm and hand tremors, which are most noticeable when at rest. In addition to this, he has noticed that his muscles have been stiffer, he feels slower, and "clumsier" when trying to do physical activity and movements.
DH is concerned that he has Parkinson's disease because his father had the condition. DH has a medical history of hypertension, constipation, and osteoarthritis. He currently takes hydrochlorothiazide 25 mg PO QAM, senna 17.2 mg QHS PRN, acetaminophen 325-650 mg PO QID PRN, and quetiapine 12.5 mg PO QHS PRN for sleep. He has no allergies to medications. He smokes 1 cigar a couple of times a year. He tries to walk at least 1 km daily. He is a retired farmer who used to engage in significant daily physical activity on the farm until his retirement 6 years ago.

On assessment, BP 128/76 mmHg; HR 68 bpm; afebrile. CBC and electrolytes within normal limits.

1. Which of the following risk factors for Parkinson's disease does DH have?
   a) Age
   b) Family history
   c) Exposure to pesticides
   d) Age, family history and pesticide exposure

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On assessment, BP 128/76 mmHg; HR 68 bpm; afebrile. CBC and electrolytes within normal limits.

2. Which of DH's drugs may potentially be contributing to his symptoms?
    a) Hydrochlorothiazide
    b) Senna
    c) Quetiapine
    d) Acetaminophen

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On assessment, BP 128/76 mmHg; HR 68 bpm; afebrile. CBC and electrolytes within normal limits.

3. Assuming DH is diagnosed with Parkinson’s Disease, which of the following medications would be most appropriate to initiate?

a) Levodopa/carbidopa
b) Entacapone
c) Pramipexole
d) Carbidopa

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On assessment, BP 128/76 mmHg; HR 68 bpm; afebrile. CBC and electrolytes within normal limits.

4. Which of the following counselling points would be LEAST appropriate to mention to DH?
   a) May cause orthostatic hypotension
   b) Take with high-protein meals
   c) Take at the same time every day
   d) May take several weeks for maximal efficacy

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CHANGE LOG

May 2025

- Slide 21: Changed to avoid in 70+

July 2025

- Formatting, grammar, spelling changes throughout
- Slide 3: Updated legend
- Slides 14, 18: Removed procyclidine
- Slide 18: Added frequency of dosing to ethopropazine
- Slide 33: Added “but requires monitoring” to clozapine

November 2025

- Significant formatting and phrasing changes made throughout
- Slide 8: Split non-motor symptoms into emotional, cognitive and autonomic
- Slide 9: Elaborated on diagnosis
- Slide 10: Added treatment
- Slide 16: Changed ‘moderate/severe’ to ‘late disease’ to match wording in guidelines
- Slide 20: Rephrased and updated benefits
- Slide 24: Added note on protein and iron interaction
- Slide 26: Elaborated on levodopa administration with protein and iron
- Slide 30: Changed dose reduction to ‘10-30%’
- Slide 36: Updated takeaway
- Slide 40: Updated answer choices

April 2026

- Slide 23: Removed brand name Sinemet® as no longer available on Canadian market
- Slide 26: Added ‘as appropriate’ to first recommendation
- Slide 30: Updated recommendations
- Slide 39: Updated question
- Slide 40: Changed question 4 for clarity and removed question 5